Analyzing E-Learning Platform Purchases using MySQL

1. Project Overview

This project focuses on analyzing an E-learning platform database using MySQL. It involves three main tables: learners, courses, and purchases, which together capture user enrollments, course details, and transaction history. SQL queries are used to explore relationships between these entities and to derive meaningful insights.

The project highlights real-world business analysis scenarios by identifying revenue trends, understanding user behavior, and evaluating course performance. Various SQL techniques such as joins and aggregations are applied to perform in-depth analysis. Overall, the study supports data-driven decision-making and helps improve business strategies.

2. Objective

- To analyze learner purchase patterns within an E-learning platform using MySQL in order to gain a clear understanding of how users interact with different courses and how frequently they make purchases.
- To calculate the total revenue generated and examine customer spending behavior across various courses so that the financial performance of the platform can be evaluated effectively.
- To identify top-performing courses and categories based on sales volume and demand, which helps in understanding which areas of learning are most popular among users.
- To study customer retention by analyzing repeat purchases and user engagement over time, thereby providing insights into user loyalty and platform effectiveness.
- To generate meaningful and actionable insights using SQL queries that can support better business decision-making and improve overall platform performance.

3. Problem Statement

- The E-learning platform currently lacks structured and meaningful insights into course purchases and overall user behavior, making it difficult to understand how users interact with the platform.
- It is challenging to manually track revenue trends and evaluate the performance of different courses and categories, leading to inefficiencies in analysis.
- There is limited visibility into which courses are in high demand and which are underperforming, affecting strategic planning and course improvement.
- Customer retention and repeat purchase behavior are not clearly understood, making it hard to measure user engagement and loyalty.

4. Attribute (Column/Feature) Details

Tables	Column Name	Description
learners	learner_id	Unique learner identifier
learners	full_name	Name of the learner
learners	country	Country of learner
courses	course_id	Unique course identifier
courses	course_name	Course title
courses	category	Course category
courses	unit_price	Price of course
purchases	purchase_id	Unique purchase ID
purchases	learner_id	Reference to learner
purchases	course_id	Reference to course
purchases	quantity	Number of courses purchased
purchases	purchase_date	Date of purchase

5. Tools and Technologies

MySQL for database creation and querying.

6. Data Preprocessing

- Database and tables were created.
- Sample data inserted into all tables.
- Duplicate constraint applied using UNIQUE key.
- Data corrections done using UPDATE queries.
- Date values modified for trend analysis.
- Ensured referential integrity using foreign keys.

7. Data Modeling

- Three tables: learners, courses, purchases.
- One-to-many relationship between learners and purchases.
- One-to-many relationship between courses and purchases.

8.Query and Output

Q1. Display each learner's total spending (quantity × unit_price) along with their country.

```
select l.learner_id,

l.full_name,

l.country,

sum(p.quantity*c.unit_price) as Tot_spend

from learners l

left join purchases p on l.learner_id=p.learner_id

left join courses c on  p.course_id=c.course_id

 group by l.learner_id;
```

Output:

learner_id	full_name	country	Tot_spen
1	Jency	USA	98969.89
2	Jeromi	India	89069.89
3	Ibrahim	UAE	87919.98
4	Maria	Spain	29000.99
5	Chen Wei	China	78090.00
6	Vennba	USA	48909.99
7	Rani	India	50059.90
8	Fatima	UAE	39009.99
9	Sandra	Spain	29000.99
10	ching chai	China	78090.00

Q2. Find the top 3 most purchased courses based on total quantity sold.

```
 select c.course_id,

 c.course_name,

 sum(p.quantity)as Total_Quantity_Sold

 from  courses c

 left join  purchases p
```

on c.course_id=p.course_id

group by c.course_id ,c.course_name order by Total_Quantity_Sold desc limit 3;

Output:

course_id	course_name	Total_Quantity_Sold
1	SQL for Beginners	3
2	Python Programming	3
3	Power BI Dashboard	3

Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.

select c.category,

sum(p.quantity*c.unit_price) as Total_Revenue ,

count(distinct p.learner_id) as Total_Students,rank() over(order by sum(p.quantity*c.unit_price) desc) as category_Rank

from courses c

left join purchases p

on c.course_id=p.course_id group by c.category ;

category	Total_Revenue	Total_Students	category_Rank
AI	314180.00	4	1
Programming	253179.70	5	2
Database	245729.97	5	3
Analytics	203029.97	5	4
Productivity	115001.98	4	5
IT & Software	NULL	0	6

Q4. List all learners who have purchased courses from more than one category.

select l.learner_id,l.full_name,count(distinct c.category)as Total_Category from learners l

join purchases p

on l.learner_id=p.learner_id

join courses c

on c.course_id=p.course_id

group by l.learner_id,l.full_name

having count(distinct c.category)>1;

Output:

learner_id	full_name	Total_Category
1	Jency	2
2	Jeromi	2
3	Ibrahim	2

Q5. Identify courses that have not been purchased at all.

```
select c.course_id ,
c.course_name
 from courses c
 left join purchases p
 on c.course_id=p.course_id   where p.course_id is null;
```

Output:

course_id	course_name
16	Cloud Computing Basics

9. Insights

Total Spending per Learner

- Learners with multiple purchases (like learner_id 1, 2, 3) show higher total spending.
- High spending is driven by purchasing multiple courses or higher-priced courses.
- Majority of learners have only one purchase → low individual contribution.
- Revenue is concentrated among a few repeat customers.
- Indicates need to increase spending from one-time users.

Top Purchased Courses

- Courses with repeated entries in purchases table have highest demand.
- Course_id 2 appears multiple times → most popular course.
- Beginner and programming-related courses are frequently purchased.
- Some courses have only one purchase → low popularity.
- Popular courses can be expanded or promoted further.

Category Revenue

- Categories with more purchases generate higher revenue (e.g., Programming, Database).
- Revenue depends on both **price and quantity sold**.
- High-priced AI courses contribute significant revenue even with fewer sales.
- Some categories have limited purchases → lower revenue contribution.
- Indicates imbalance in category performance.

Repeat Customers

- Learners 1, 2, and 3 have multiple purchases → repeat customers.
- Repeat customers contribute more to total revenue.
- Majority of learners are one-time buyers.
- Retention rate is relatively low.
- Opportunity to improve engagement and loyalty programs.

Monthly Revenue Trend

- Most purchases are concentrated in April .
- Few records exist for other months → uneven distribution.
- Revenue increases slightly with added transactions mid-month.
- No clear long-term trend due to limited monthly variation.
- More consistent data needed for better trend analysis.

Country-wise Revenue

- India, USA, and UAE have multiple learners contributing to revenue.
- Countries with more learners generate higher revenue.
- Revenue distribution is uneven across countries.
- Some countries have only one learner → low contribution.
- Helps identify key markets for targeting.

10. Analysis

Descriptive Analysis

- The total number of learners in the dataset shows that the platform has a growing user base, with multiple new learners added beyond the initial entries, indicating steady expansion.
- The total number of courses available covers multiple categories such as Database, Programming, Analytics, Productivity, and AI, which reflects a diverse learning platform.
- The total number of purchases made indicates active user participation, with several learners making transactions across different dates, especially concentrated in April.
- The total revenue generated is calculated using quantity and unit price, showing that high-priced courses (like AI and advanced courses) significantly contribute to overall revenue.
- Category-wise sales reveal that categories such as Programming and Database have higher purchase frequency, while some categories have relatively lower engagement.

Diagnostic Analysis

- Certain courses sell more because they appear multiple times in the purchases table (for example, course_id 2), indicating strong learner interest and demand for those topics.
- Comparison of categories shows that revenue is not evenly distributed, as some categories generate more income due to either higher prices or more frequent purchases.
- Customer purchase frequency analysis reveals that only a few learners (such as learner_id 1, 2, and 3) make repeated purchases, while most learners buy only once.
- Low-performing categories are identified by fewer purchase records, suggesting either lack of demand, lower visibility, or less relevant content.
- Demand patterns indicate that most transactions are concentrated within a specific time period (April), showing limited variation in purchasing behavior across months.

Predictive Analysis

- Future revenue trends are expected to grow if the current pattern of repeat purchases and course expansion continues, especially in high-performing categories.
- Demand forecasting suggests that courses already popular (like those repeatedly purchased) will continue to attract more learners in the future.
- Customer growth prediction indicates that as more learners are added, overall purchases and revenue are likely to increase proportionally.
- Seasonal purchase trends cannot be fully determined due to limited variation in months, but current data suggests peak activity during certain periods.

- High-demand category prediction shows that categories like Programming and Database are likely to remain dominant based on current purchase behavior.

Prescriptive Analysis

- It is recommended to promote and expand popular courses (such as frequently purchased ones) by adding advanced levels or bundled offers to maximize revenue.
- Weak or low-performing categories should be improved by updating course content, enhancing quality, or increasing marketing efforts to boost engagement.
- Discounts and offers should be applied strategically during peak periods or on less popular courses to increase sales and attract more learners.
- Repeat customers should be targeted with loyalty programs, personalized recommendations, or exclusive benefits to increase retention and spending.
- Marketing strategies should focus more on high-revenue countries and high-demand categories to optimize growth and improve overall platform performance.

11. Conclusion

- The project successfully analyzed the E-learning database by combining learners, courses, and purchases data, providing a complete understanding of platform activity and user interactions.
- It identified clear revenue and sales patterns, showing that a few high-demand courses and repeat customers contribute significantly to overall revenue.
- The analysis highlighted top-performing courses and categories (such as frequently purchased courses like course_id 2), helping understand learner preferences and market demand.
- Customer behavior analysis revealed low retention overall, with only a few learners making multiple purchases, indicating the need for better engagement strategies.
- Overall, the project demonstrates how SQL techniques (joins, aggregations, grouping) can be effectively used to generate data-driven insights and support business growth decisions.